

Manifold for Type 1 Diabetes – Restoration

Research-partnership brief — β -cell function preservation, regeneration, and replacement-therapy analytics

1 · THE CLINICAL PROBLEM

T1D does not end at diagnosis — it begins a decades-long contest for residual and restored β -cell function. Three clinical paradigms are now converging:

- **Preservation.** Residual β -cell mass at diagnosis predicts long-term complications, glycemic stability, and insulin requirement. Current practice treats the honeymoon phase as luck; a handful of agents (teplizumab extension, low-dose ATG, verapamil, GLP-1 combinations) measurably slow C-peptide decay but response is heterogeneous.
- **Regeneration.** Pharmacologic β -cell proliferation is no longer hypothetical — DYRK1A inhibitors (harmine-class), GLP-1 / DYRK1Ai combinations, and senolytics have moved into human studies. The readout is slow and multi-signal; classical trial statistics under-exploit the data.
- **Replacement.** Stem-cell-derived islet therapies (Vertex VX-880, Sernova Cell Pouch, ViaCyte/CRISPR) have produced insulin-independence in early-phase patients. Each trial has small n and immense per-patient signal; graft-survival and immune-escape dynamics are the bottleneck.

What today's analytic practice does not do well:

- Separate heterogeneous treatment effect (who responds to teplizumab vs. verapamil vs. combo) from noise at realistic sample sizes.
- Treat post-infusion metabolic trajectories (C-peptide, CGM variability, inflammatory panels) as a *joint dynamical system* rather than parallel endpoints.
- Rationalize combination-therapy design (β -cell proliferation + immune tolerance + metabolic support) on a causal graph rather than by empirical stacking.

Vertex VX-880's first cohort achieving insulin-independence is a proof-of-principle, not a finished product. The question is no longer "can β -cell function be restored?" — it is "for whom, with what adjuncts, and for how long?" That question is causal-inference-shaped.

2 · WHAT MANIFOLD BRINGS

Manifold is a causal-intelligence platform for detecting structural fragility and regime transitions in complex multi-signal systems. Three engine capabilities anchor the restoration configuration:

1. **Causal discovery (Spirtes / PC algorithm) on multi-omic + clinical-trial data.** Recovers the directed structure between genotype, autoantibody profile, drug exposure, and C-peptide / CGM response — the substrate for principled heterogeneous-treatment-effect inference.
2. **Criticality estimators feeding a composite fragility score (ΩF).** Configured for post-intervention dynamics: honeymoon-phase hazard, graft-compromise early-warning, combo-therapy response separation.
3. **Pearl do-calculus / interdiction.** Frame combination-therapy design as max-attenuation intervention on the inferred causal graph — rational ordering of β -cell-proliferative, immune-tolerogenic, and metabolic adjuncts instead of empirical stacking.

3 · ΩF PILLAR MAPPING - RESTORATION PHYSIOLOGY

ΩF PILLAR	RESTORATION READING
I — Interaction	Drug $\times$ genotype $\times$ autoantibody-profile interactions for combo response; graft $\times$ host-immune compatibility
R — Reserve	Residual β -cell mass (C-peptide AUC on MMTT); post-infusion graft functional mass; stress-test response under mixed-meal or glucagon challenge
J — Junction	Immune–metabolic coupling: cytokine-panel $\leftrightarrow$ C-peptide transfer entropy; how continuing autoimmunity interacts with regenerated or transplanted β -cells
C — Cascade	β -cell senescence and ER-stress cascades; DYRK1A $\rightarrow$ cell-cycle-progression cascade; inflammatory escalation after graft exposure
T — Temporal	Hazard of honeymoon-end, graft-failure, or response-loss given current trajectory; time-to-restored-independence

4 · CRITICALITY ESTIMATOR CONFIGURATION

Restoration data is small- n , longitudinally rich, and heterogeneous-in-response. The criticality layer is configured accordingly:

ESTIMATOR	WHAT IT CATCHES	PILLAR FEED
Nonlinear mixed-effects C-peptide decay models	Per-subject honeymoon-phase prognosis; subject-level deviation from cohort decay curve	R, T
Heterogeneous-treatment-effect learners (Causal Forest, S / T / X-learners, BART)	Who responds to which drug or combination, under realistic trial sample sizes	I, C
Gaussian process regression on sparse post-infusion trajectories	Graft-function estimation and uncertainty with $n < 20$ and irregular follow-up	R, T
Bayesian online change-point (BOCPD) on CGM + inflammatory panels	Graft-compromise and honeymoon-end detection with calibrated uncertainty	J, T

Complementary estimators available in the same layer for v2:

- **Transfer entropy** between inflammatory cytokine panels and C-peptide / CGM signals — quantifies the immune-metabolic coupling (J) that drives graft rejection and regenerative failure.
- **Variance-based early-warning signals** on high-frequency CGM windows in the first 30 days post-infusion — leading indicator of graft-compromise before mean-function drops.

5 · MINIMUM VIABLE COLLABORATION (V1)

Smallest slice that produces something a principal investigator or trial steward will take seriously.

- **Dataset option A — preservation.** Retrospective teplizumab / ATG / verapamil trial data with serial C-peptide AUC, MMTT, autoantibody titers. Joslin T1D Medalist honeymoon-cohort as long-horizon validation.
- **Dataset option B — replacement.** Published or collaborator-provided phase 1/2 islet-cell-therapy post-infusion trajectories (Vertex VX-880 cohort-level summaries or partner-site raw data; Sernova Cell Pouch clinical data).
- **Access path.** Breakthrough T1D or NIDDK HIRN-sponsored DUA with a trial-adjacent PI or consortium.
- **Deliverable.** Per-subject Ω F trajectory with mixed-effects + HTE + GP agreement, benchmarked against the trial's published biostatistical analysis. For preservation: 6 / 12 / 24-month C-peptide hazard. For replacement: graft-function hazard and combo-response classification.
- **Success criteria.** HTE recovery on a held-out trial arm matching or exceeding the pre-specified biostatistical analysis; identification of at least one interpretable responder / non-responder subgroup decomposable along Ω F pillars.
- **Timeline.** 12–14 weeks from data access.
- **Out of scope v1** (reserved for v2 / prospective arm): live CGM ingestion from open-loop pumps, bespoke immunologic assays, multi-site prospective design.

6 · WHY THIS, WHY NOW

- **Replacement has crossed the proof-of-principle threshold.** Insulin-independence in early-phase Vertex VX-880 patients is a stage-change for the field. The next five years are about *generalization* — who responds, with which adjuncts — which is an analytic problem before it is a clinical one.
- **Pharmacologic regeneration is clinically tractable.** DYRK1A inhibitor programs and GLP-1 / DYRK1Ai combinations are in human studies; their readout is multi-signal and slow, exactly the regime where composite-fragility framing outperforms endpoint-by-endpoint analysis.
- **Preservation has approved agents.** Teplizumab post-diagnosis, verapamil, low-dose ATG, and GLP-1 adjuncts are all actionable today; the analytic gap is matching patient to drug, not discovering new drugs.
- **Funding lines are open.** Breakthrough T1D Cures program, Helmsley Charitable Trust "Restoring Insulin Production" stream, and NIH NIDDK Human Islet Research Network (HIRN) consortium all have active RFPs in this space.
- **The platform is ready.** Manifold's causal, criticality, and interdiction engines are production-grade; configuring them for the restoration domain is bounded, well-scoped work.

Partnership with a β -cell-regeneration, islet-transplant, or preservation-trial PI (or cohort steward — Joslin, Mount Sinai Stewart Lab, Vertex trial collaborator, or HIRN consortium member) for:

1. Retrospective trial or registry data access under their existing Data Use Agreement.
2. Clinical co-authorship on the heterogeneous-treatment-effect / graft-survival benchmark paper.
3. Co-application to Breakthrough T1D Cures, Helmsley Restoring Insulin Production, or NIH NIDDK HIRN for the v2 prospective arm.

In exchange, Apex Analytica brings the causal-inference engine, the Ω F framework, and the software team. Apex retains commercial rights to the platform; research IP from the joint work is shared per standard academic-industry terms.